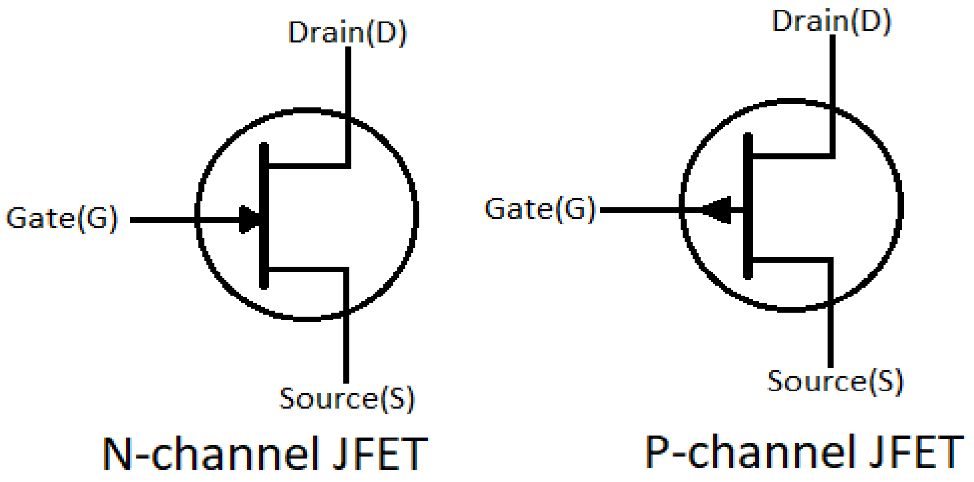
FET Parameters and MOSFET

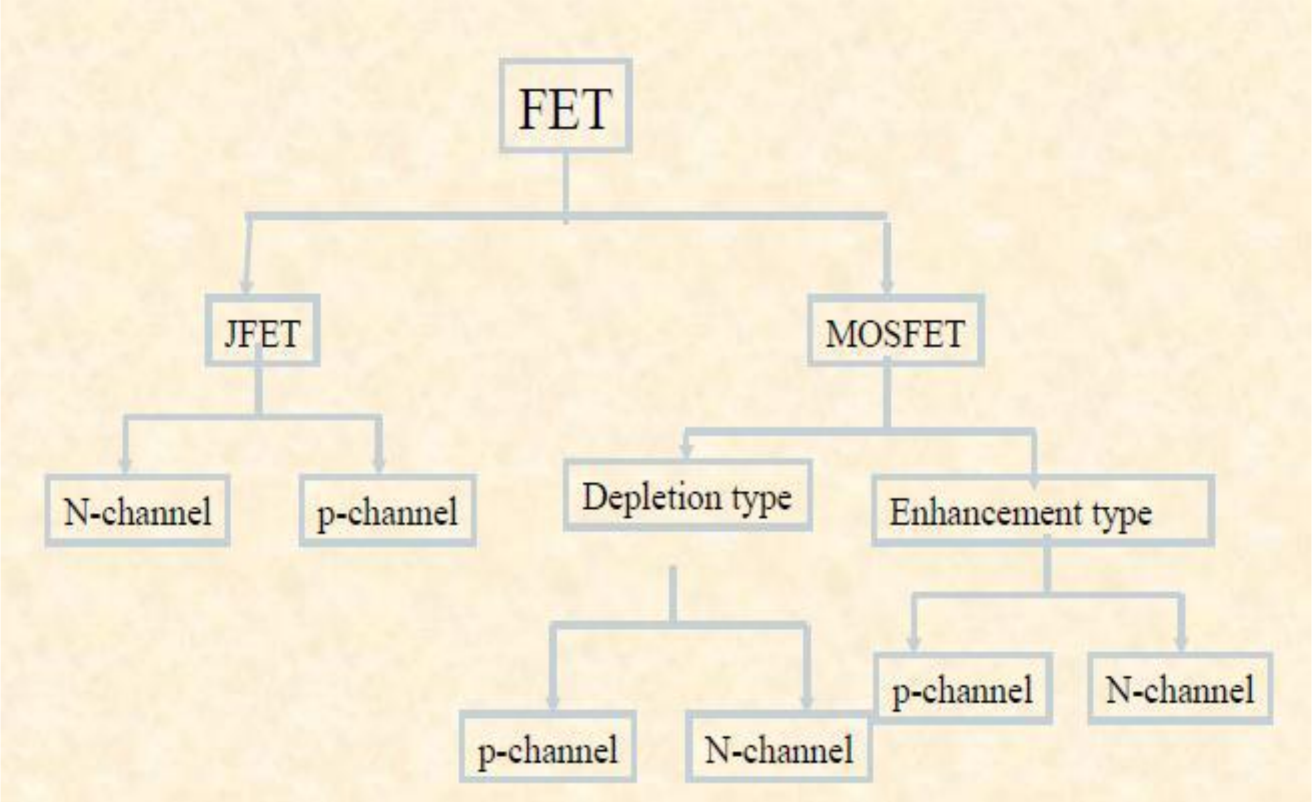
Introduction

- It is a semiconductor device whose operation depends on the control of current by an electric field.
- The current conduction in FET is due to majority charge carriers only. Therefore, FETs are known as unipolar devices.
- BJT is current control device and FET is a voltage control device.



Classification of FET

- FET
  - Junction field effect transistors (JFET)
  - Metal oxide semiconductor field effect transistor (MOSFET)



- Based on the construction JFETs are of two types
  - N Channel FET
  - P Channel FET

JFET characteristics

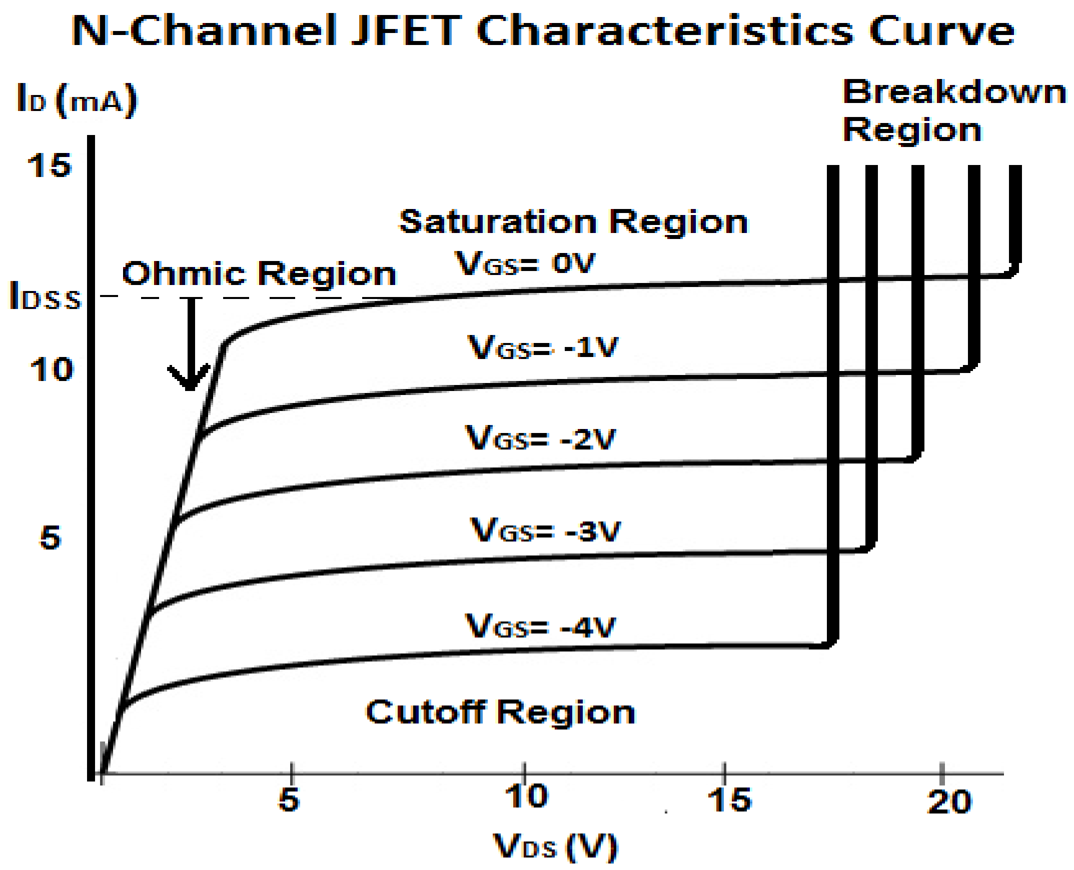


Fig. Output or drain characteristics

control variable

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

constants

Vgs	I_D
0	I_{DSS}
0.3Vp	$I_{DSS}/2$
0.5Vp	$I_{DSS}/4$
Vp	0 mA

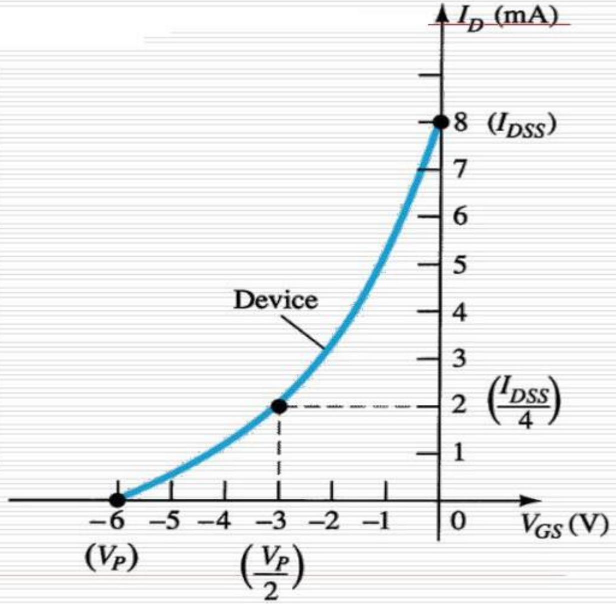


Fig. Transfer characteristics

JFET Parameters

- **DC Drain resistance**: Defined as ratio of drain to source Voltage (V_{DS}) to drain current (I_D). It is also called static or Ohmic Resistance

Mathematically it can be represented as

$$R_{DS} = \frac{V_{DS}}{I_D}$$

- **AC Drain resistance ($\boldsymbol{r_d}$):** Defined as the resistance between drain to source when JFET is operating in Pinch off Region or saturation region. AC drain resistance is given by the equation

$$r_d = \frac{\Delta\, V_{DS}}{\Delta\, I_D} \text{ at constant } V_{GS}$$

- **Transconductance ($\boldsymbol{g_m}$):** It is given by the ratio of small change in drain current to the corresponding change in the Gate to source Voltage V_{GS}. Also known as Forward Transmittance

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} \text{ at constant } V_{DS}$$

JFET Parameters

- **Amplification factor (μ)**: Defined as ratio of change in drain to source Voltage (ΔV_{DS}) to the corresponding change in the gate to source voltage V_{GS}.

  Mathematically it can be represented as

$$\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}}$$

- **Relation between μ, $\boldsymbol{r_d}$, and $\boldsymbol{g_m}$:**

$$\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}} = \frac{\Delta V_{DS}}{\Delta I_D} \times \frac{\Delta I_D}{\Delta V_{GS}}$$

$$= r_d \times g_m$$

Exercise-1:Prove that $g_m = g_{mo}\left[1 - \frac{V_{GS}}{V_p}\right]$?

- **Solution:**

$$I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_p}\right]^2 \quad \text{—— (1).}$$

$$\Rightarrow \frac{dI_D}{dV_{GS}} = \frac{-2 \cdot I_{DSS}}{V_p}\left(1 - \frac{V_{GS}}{V_p}\right) \quad \text{(Differentiating both side w.r.t } V_{GS}\text{)}$$

$$\Rightarrow g_m = \frac{-2 I_{DSS}}{V_p}\left[1 - \frac{V_{GS}}{V_p}\right].$$

when $V_{GS} = 0$, the above eqn became.

$$\boxed{g_m = \frac{-2 I_{DSS}}{V_p} = g_{mo}}$$

$$\therefore \boxed{g_m = g_{mo}\left[1 - \frac{V_{GS}}{V_p}\right]}$$

Exercise-2:Prove that $g_m=\frac{2}{|V_p|}\sqrt{I_{DSS}\times I_{DS}}$?

- **Solution:**

Ans: From the shockly's eqn we know.

$$I_{DS} = I_{DSS}\left[1-\frac{V_{GS}}{V_p}\right]^2 \quad \text{—— (1)}$$

Differentiating bothside w.r.t V_{GS}.

$$\frac{dI_{DS}}{dV_{GS}} = -\frac{2I_{DSS}}{V_p}\cdot\left[1-\left(\frac{V_{GS}}{V_p}\right)\right]$$

$$\Rightarrow g_m = -2\cdot\frac{I_{DSS}}{V_p}\left[1-\left(\frac{V_{GS}}{V_p}\right)\right] \quad \text{—— (2)}$$

From eqn (1) we have

$$\frac{I_{DS}}{I_{DSS}} = \left[1-\left(\frac{V_{GS}}{V_p}\right)\right]^2$$

$$\Rightarrow \left[1-\left(\frac{V_{GS}}{V_p}\right)\right] = \sqrt{\frac{I_{DS}}{I_{DSS}}} \quad \text{—— (3)}$$

Putting this value in eqn (2) we have.

$$g_m = -\frac{2I_{DSS}}{V_p}\sqrt{\frac{I_{DS}}{I_{DSS}}}$$

$$\boxed{g_m = \frac{2}{|V_p|}\sqrt{I_{DSS}\cdot I_{DS}}}$$

JFET Vs BJT: Summary

JFET		BJT
$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$	$\Leftrightarrow$	$I_C = \beta I_B$
$I_D = I_S$	$\Leftrightarrow$	$I_C \cong I_E$
$I_G \cong 0\text{ A}$	$\Leftrightarrow$	$V_{BE} \cong 0.7\text{ V}$

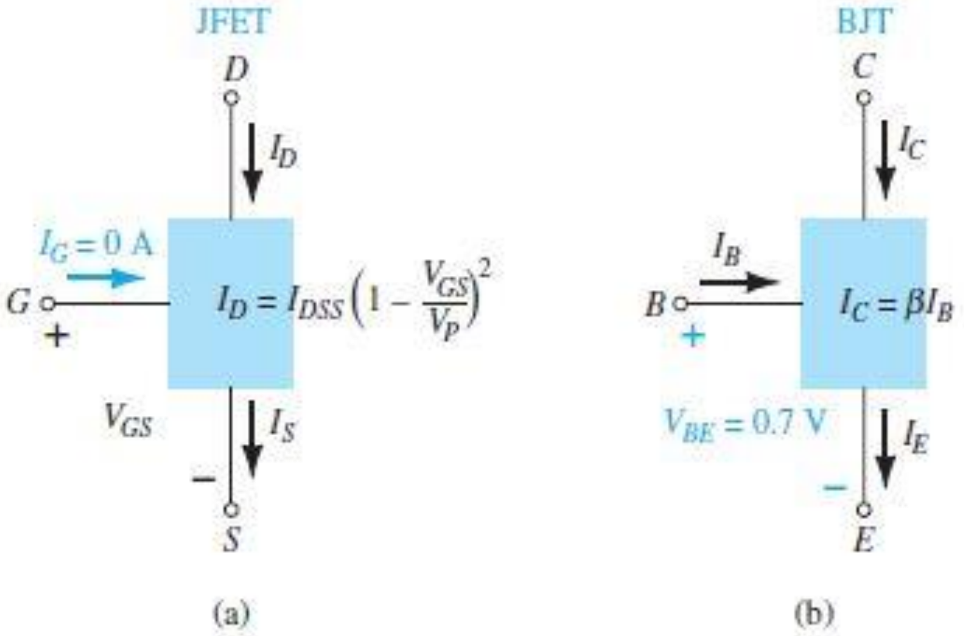


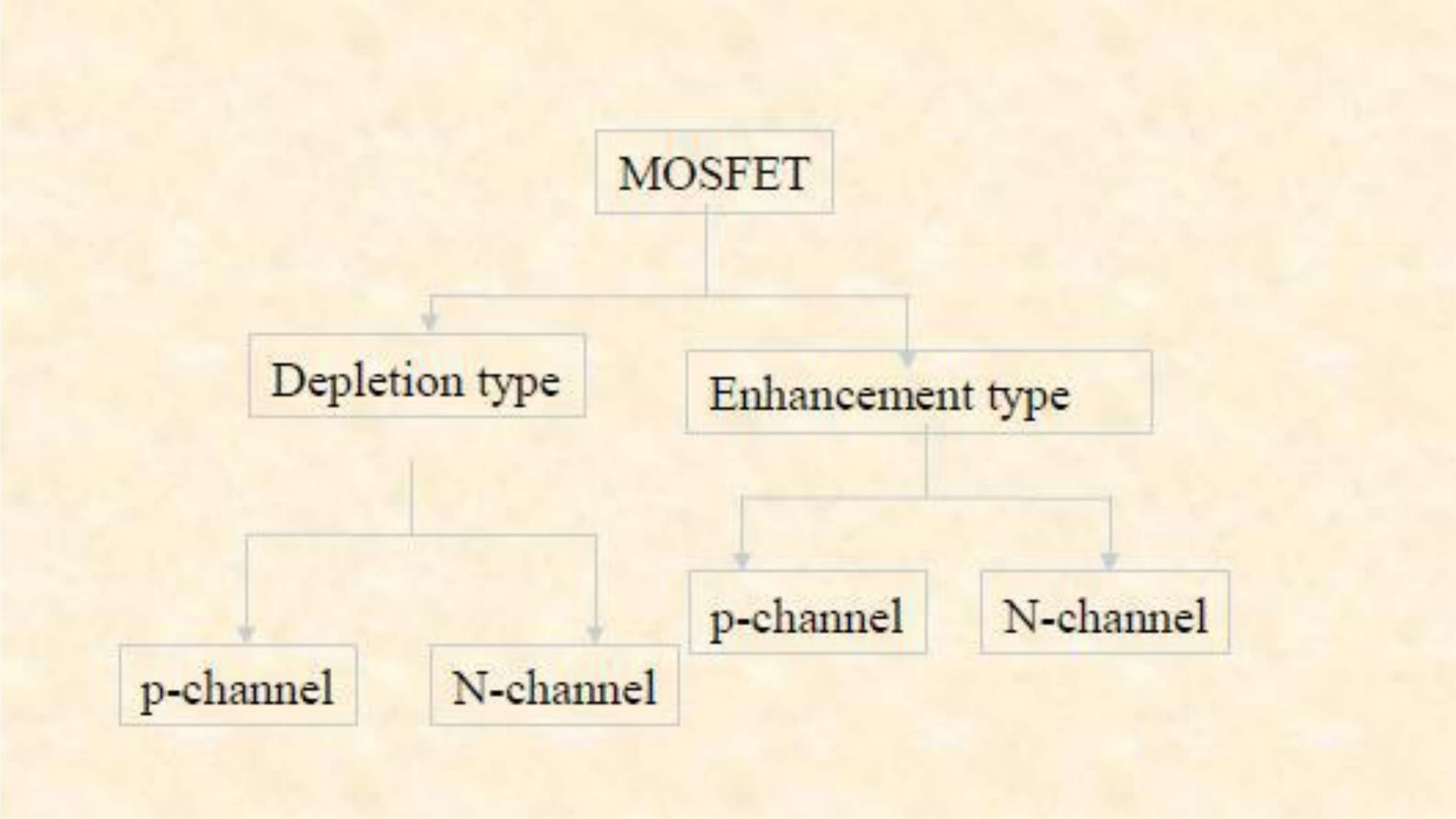
Fig. (a) JFET versus (b) BJT.

Metal oxide semiconductor field effect transistor (MOSFET)

Introduction

- MOFET is a three terminal semiconductor device with gate insulated from the channel.
- It is also called as insulated gate FET (IGFET).
- MOSFETs uses a metal gate electrode (instead of p-n junction in JFET), separated from the semi conductor by an Insulating thin layer SiO2 to modulate the resistance of the conduction channel.
- The drawback of FET is the gate of the FET must be reverse bias for it's operation which is known as depletion mode i.e. it uses an input voltage to reduce the channel size from it's zero bias size.
- MOSFET can operate both in depletion as well as enhancement mode.
- The input impedance of the MOSFET is higher than FET.

MOSFET types



D-MOSFET

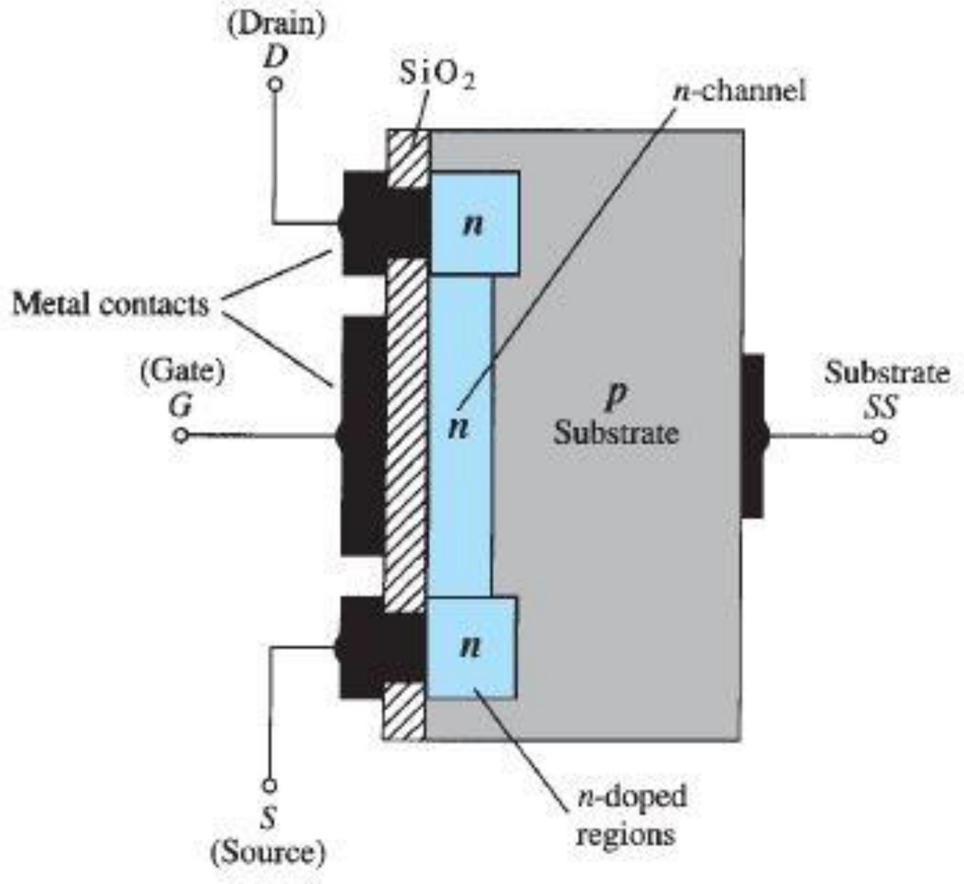


Fig. Construction of n-channel depletion-type MOSFET.

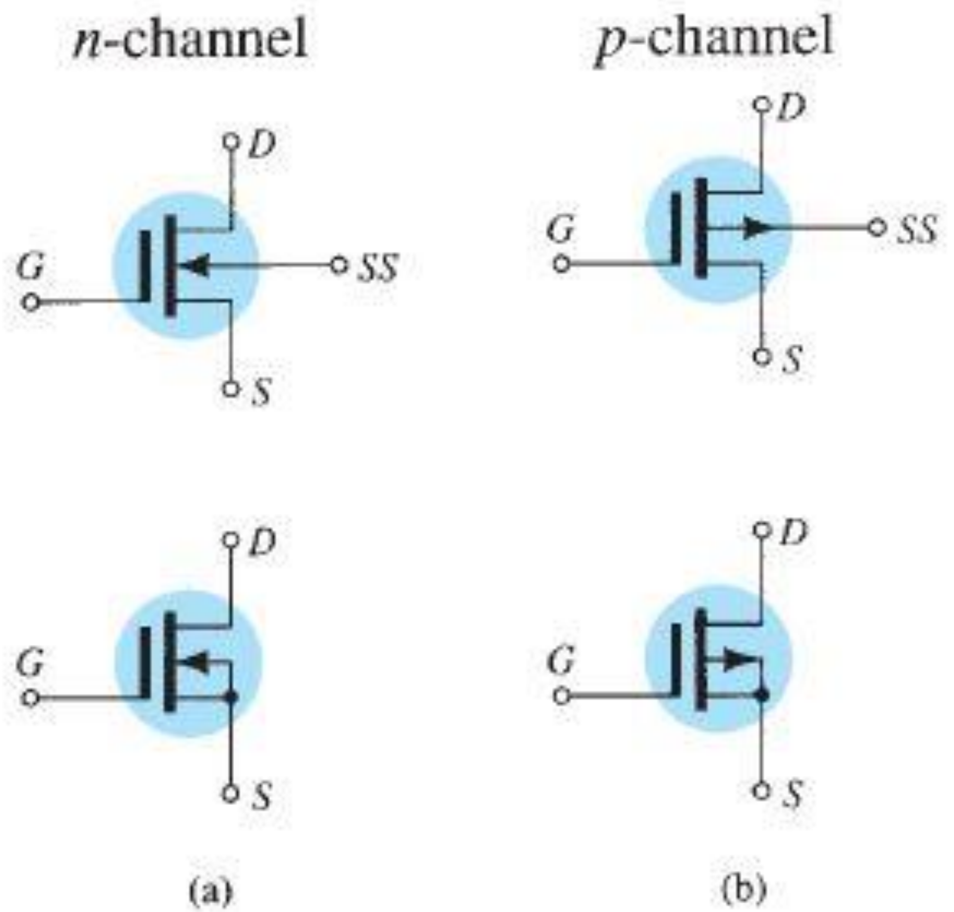


Fig. Symbols for: (a) n-channel depletion-type MOSFETs and (b) p-channel depletion-type MOSFETs.

D-MOSFET operation

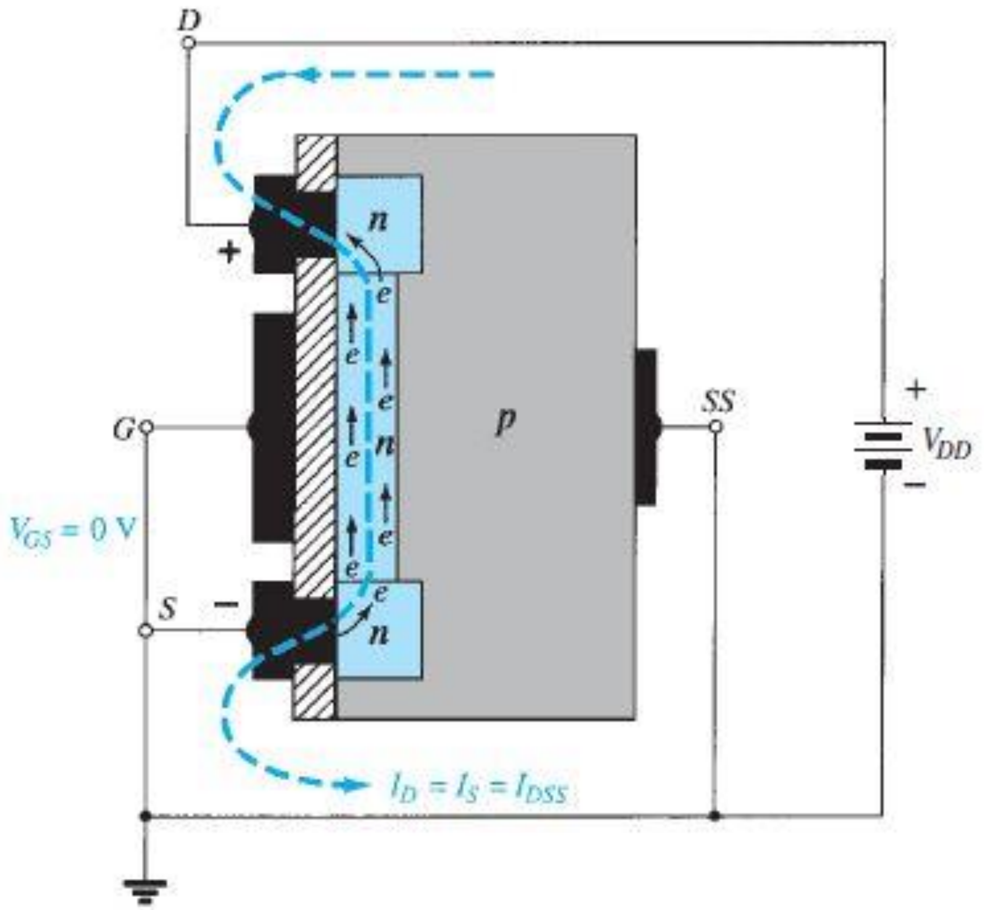


Fig. n-Channel depletion-type MOSFET with V_{GS} = 0 V and applied voltage V_{DD} .

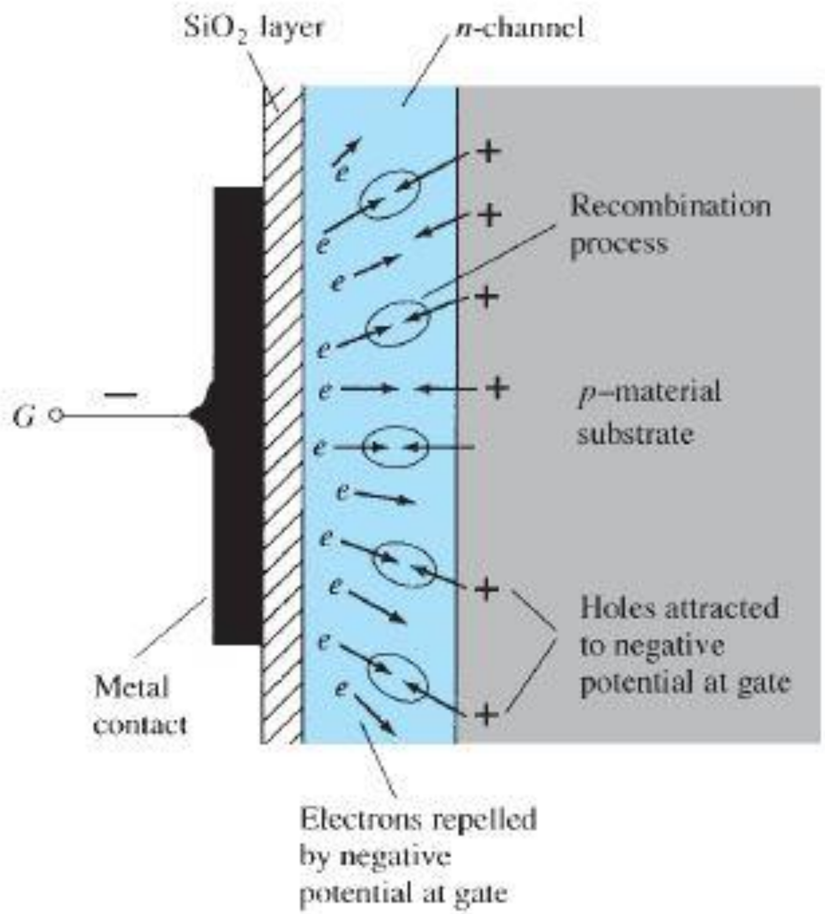


Fig. Reduction in free carriers in a channel due to a negative potential at the gate terminal.

D-MOSFET characteristics

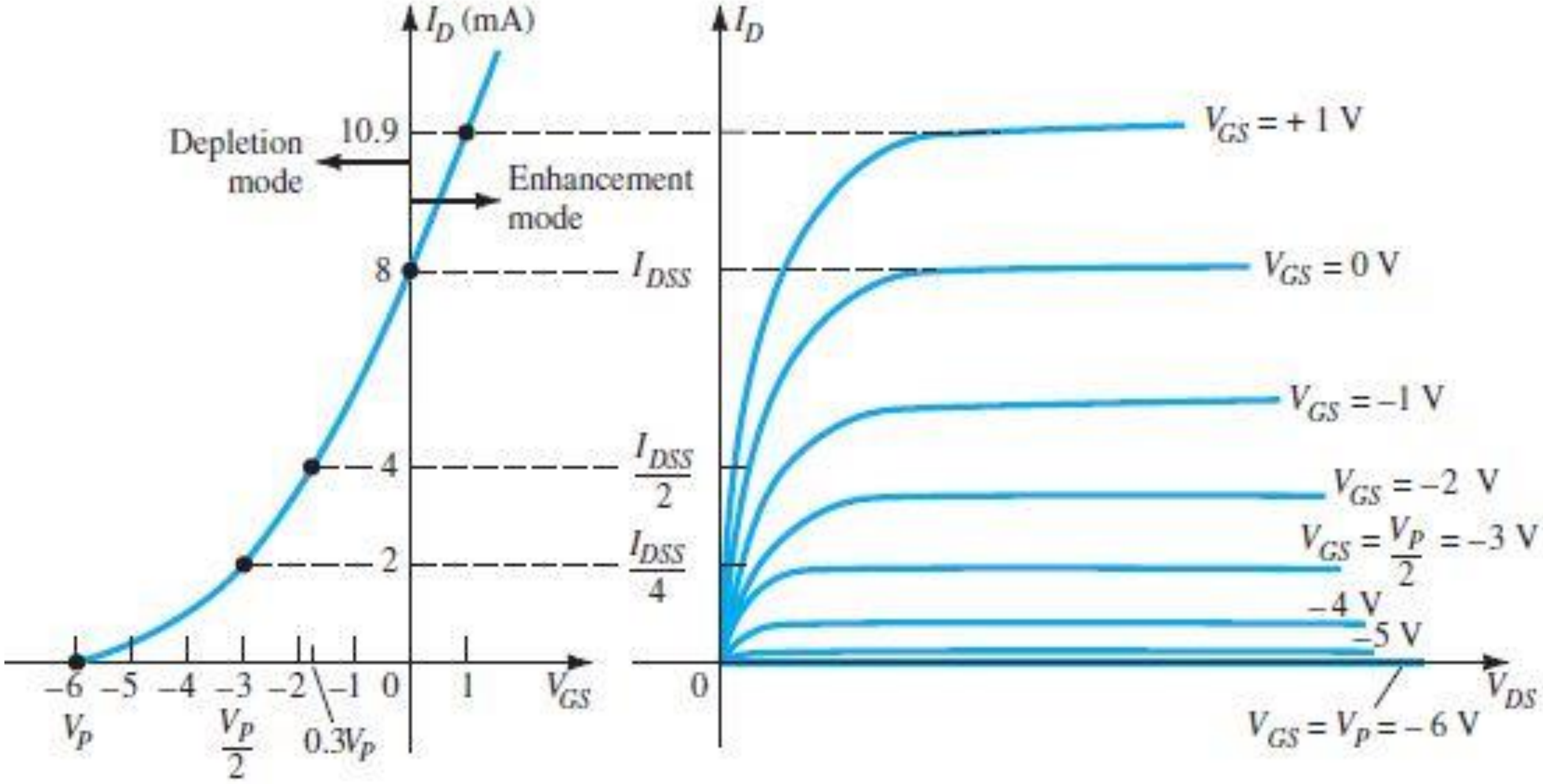


Fig. Drain and transfer characteristics for an n-channel depletion-type MOSFET.

E-MOSFET

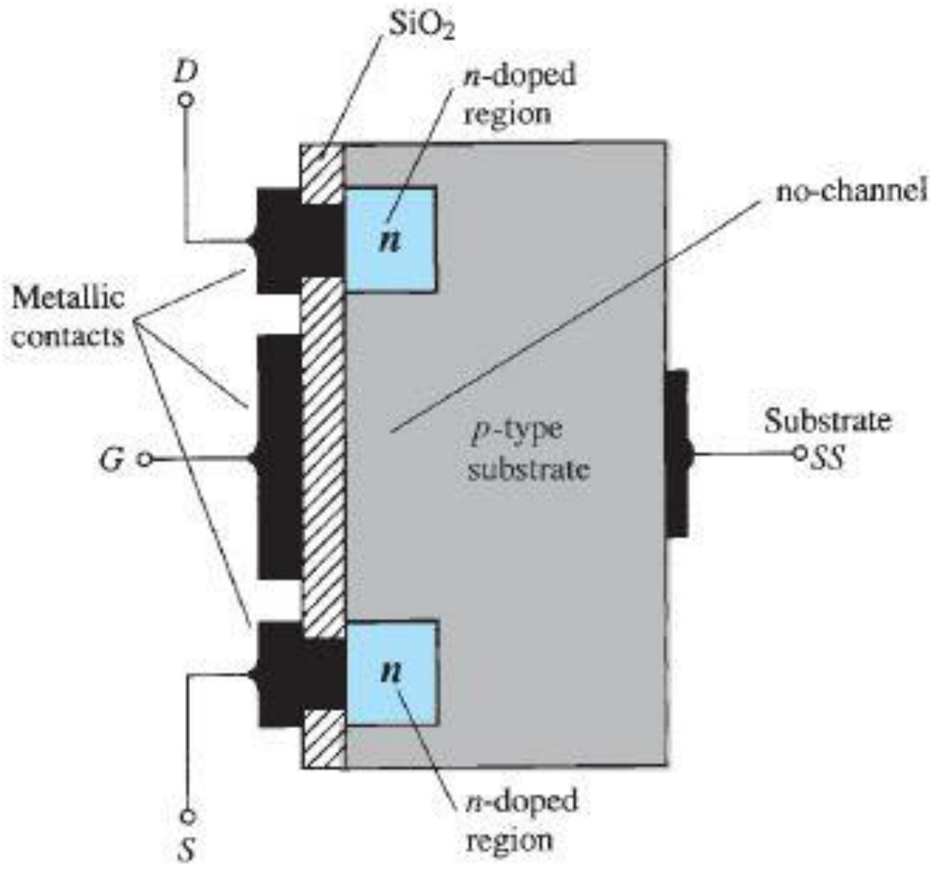


Fig. Construction of n-channel enhancement-type MOSFET.

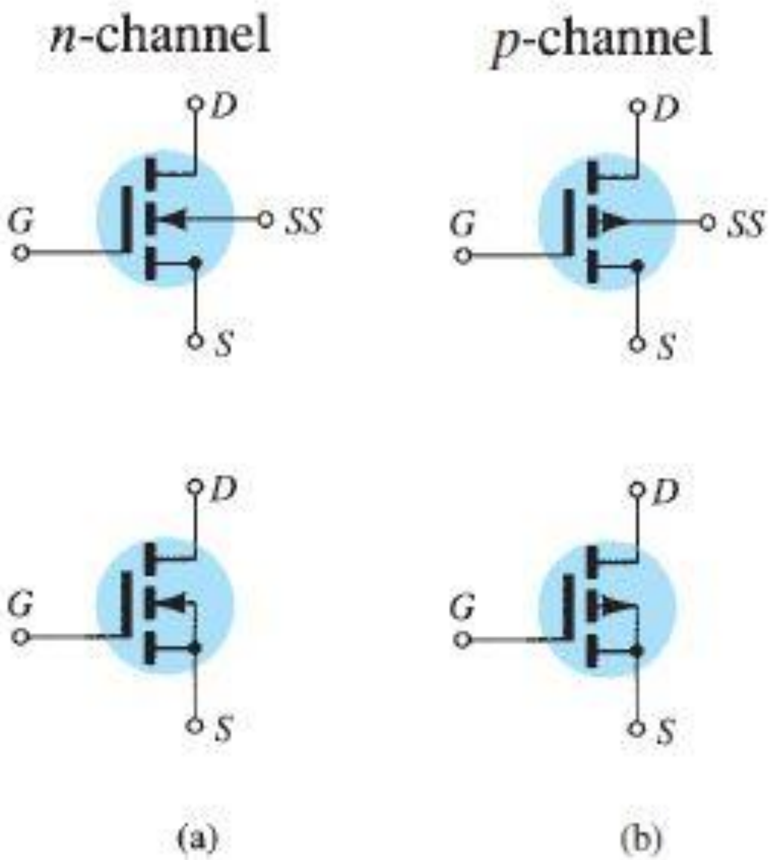


Fig. Symbols for: (a) n-channel enhancement type-type MOSFETs and (b) p-channel enhancement -type MOSFETs.

E-MOSFET operation

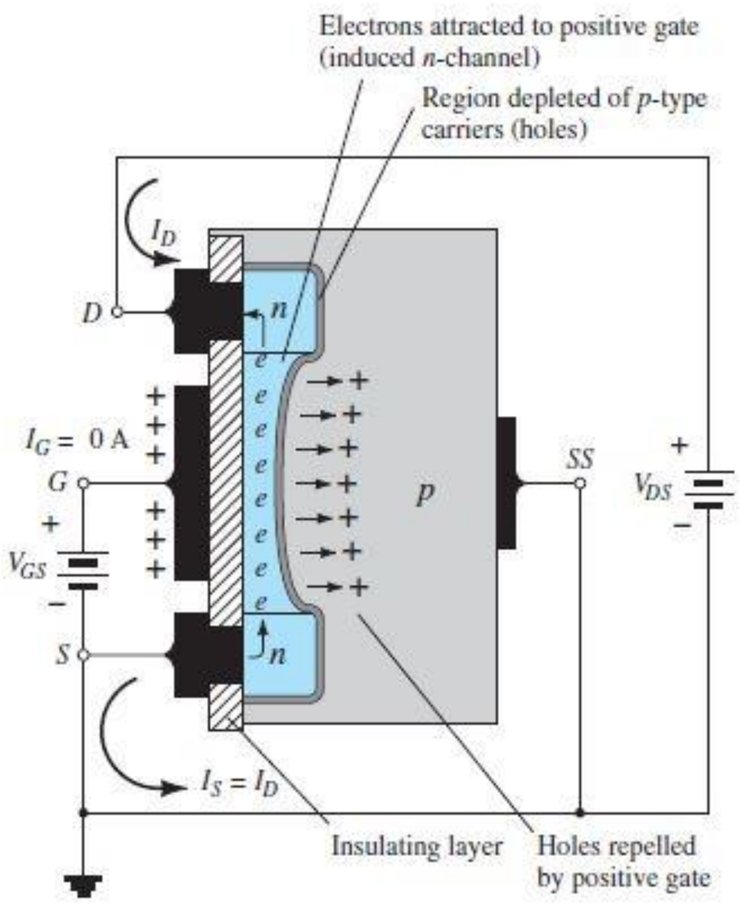


Fig. Channel formation in the n-channel enhancement-type MOSFET.

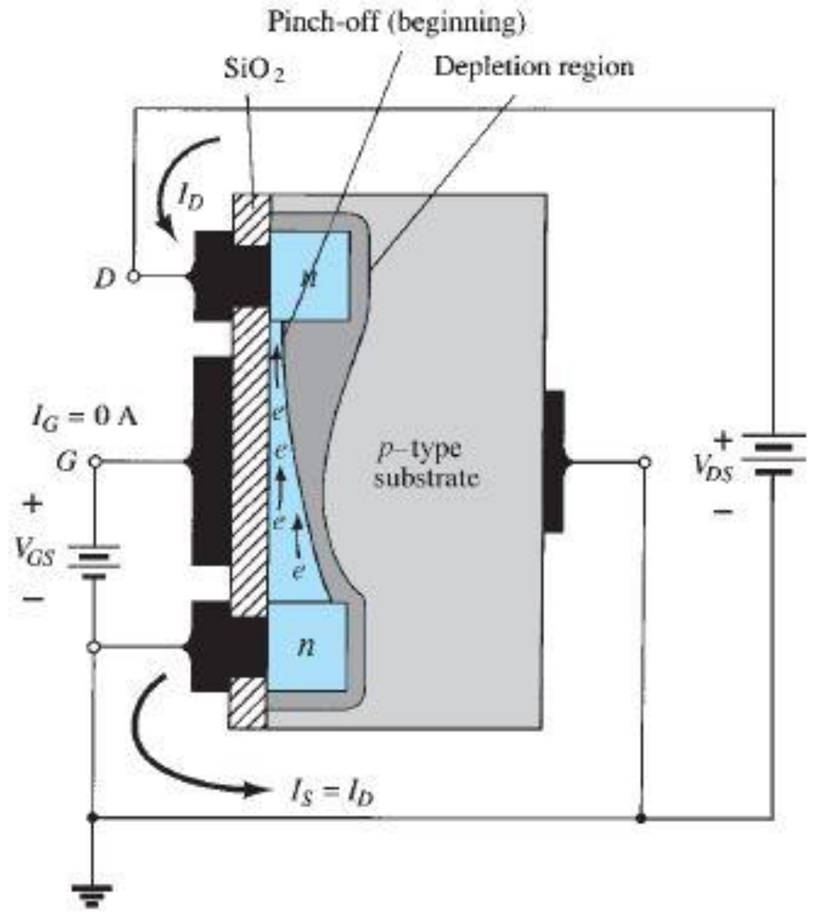


Fig. Change in channel and depletion region with increasing level of V_{DS} for a fixed value of V_{GS} .

E-MOSFET characteristics

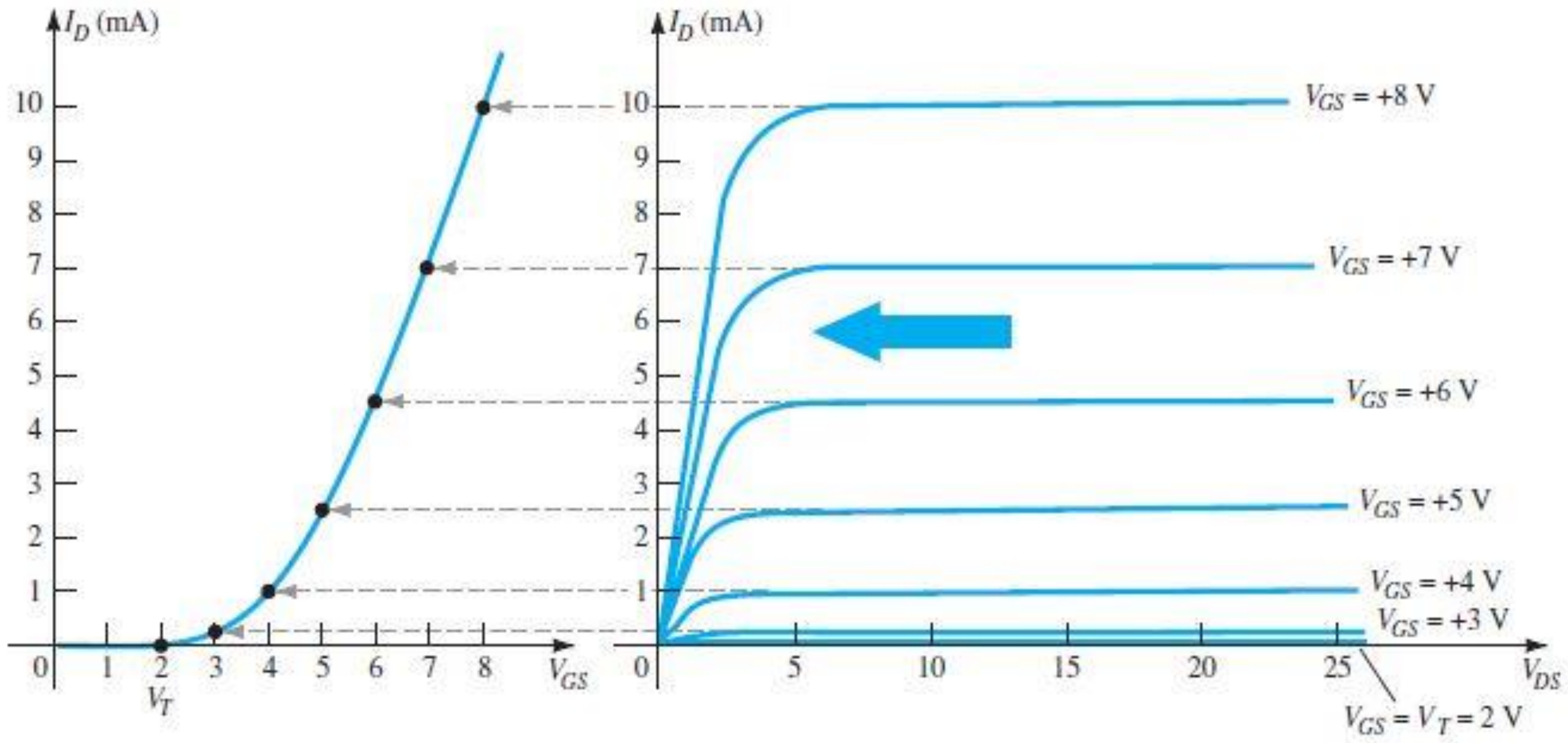


Fig. Transfer characteristics for an n-channel enhancement-type MOSFET from the drain characteristics

Thank you